Rubrik CDM Pre-Upgrade Compatibility Assessment Tool

Customer Setup & Usage Guide

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1. Overview

This tool performs an automated pre-upgrade compatibility assessment across all Rubrik CDM clusters connected to your RSC (Rubrik Security Cloud) or RSC-P (Proactive) instance. It identifies potential upgrade blockers, compatibility issues, and risks before performing a CDM version upgrade.

The tool uses two data collection methods:

- RSC GraphQL API — cluster metadata, workload inventory, SLA policies, compatibility data
- CDM Direct REST API — detailed checks including live mounts, system status, DNS/NTP, archive and replication configuration (connects directly to CDM cluster node IPs using the same service account)

The assessment generates interactive HTML dashboards, CSV spreadsheets, and JSON data files that can be shared with your Rubrik SE for upgrade planning.

2. Prerequisites

Required

Requirement	Details
Python	3.9 or higher
Network Access	HTTPS (port 443) to your RSC/RSC-P instance
RSC Service Account	With appropriate permissions (see Section 3)

Recommended

Requirement	Details
CDM Network Access	HTTPS (port 443) to CDM cluster node IPs for Direct API checks. If not reachable, the tool falls back to GraphQL-only mode automatically.

3. RSC Service Account Setup

Creating the Service Account

1. Log in to Rubrik Security Cloud (RSC)
2. Navigate to Settings → Service Accounts
3. Click "Create Service Account"
4. Name it (e.g., "cdm-upgrade-assessment")
5. Assign a role (see permissions below)
6. Save and copy the credentials:

You will need these three values:

- Client ID
- Client Secret
- Access Token URI

Minimum Permissions (Read Only)

The tool works with a Read Only Admin role at minimum. This provides:

Permission	Used For
View clusters	Cluster discovery and metadata
View workloads	Hypervisor, database, host inventory
View SLA domains	SLA policy details
View activities	Running/queued job counts

Recommended Permissions (Full Data)

For the most complete assessment, add the "Upgrade clusters" (UPGRADE_CLUSTER) permission. This enables:

Permission	Used For
Upgrade clusters	Release details from Rubrik Support Portal, recommended upgrade versions, available upgrade paths, official compatibility matrix link

ℹ️ **NOTE:** Without the "Upgrade clusters" permission, the tool uses built-in static data for EOS dates, upgrade paths, and recommendations. This data is accurate but may not reflect the very latest Rubrik releases.

CDM Direct API Access

The RSC service account credentials are also used for CDM Direct API access. The tool authenticates to each CDM cluster node using the same Client ID and Secret. No additional CDM-level configuration is needed.

ℹ️ **NOTE:** Some CDM REST API endpoints (service account management, API token enumeration) are blocked for all token-based authentication by CDM design. The tool handles this gracefully and flags these as manual verification items in the report.

4. Installation

Unzip the provided rubrik-cdm-upgrade-assessment.zip to any directory, then follow the instructions for your operating system.

4.1 macOS

Install Python (if needed):

```
brew install python@3.11
```

Run the setup script:

```
cd rubrik-cdm-upgrade-assessment
chmod +x setup.sh run.sh
./setup.sh
```

Configure credentials:

```
cp .env.example .env
nano .env   # or: open -e .env
```

4.2 Linux (Ubuntu/Debian)

Install Python (if needed):

```
sudo apt update
sudo apt install python3 python3-venv python3-pip
```

Run the setup script:

```
cd rubrik-cdm-upgrade-assessment
chmod +x setup.sh run.sh
./setup.sh
```

Configure credentials:

```
cp .env.example .env
nano .env
```

4.3 Linux (RHEL/CentOS)

Install Python (if needed):

```
sudo yum install python3 python3-pip
```

Run the setup script:

```
cd rubrik-cdm-upgrade-assessment
chmod +x setup.sh run.sh
./setup.sh
```

Configure credentials:

```
cp .env.example .env
vi .env
```

4.4 Windows

Install Python from https://python.org/downloads

! **IMPORTANT:** During Python installation, check the box "Add Python to PATH".

Open PowerShell and run:

```
cd C:\rubrik-cdm-upgrade-assessment
python -m venv .venv
.venv\Scripts\Activate.ps1
pip install --upgrade pip
pip install -r requirements.txt
```

```
copy .env.example .env
notepad .env
```

ℹ️ **NOTE:** If you get an execution policy error, run:

Set-ExecutionPolicy -ExecutionPolicy RemoteSigned -Scope CurrentUser

5. Configuration

Edit the .env file with your RSC credentials:

```
# RSC Service Account Credentials (REQUIRED)
RSC_CLIENT_ID=your-client-id-here
RSC_CLIENT_SECRET=your-client-secret-here
RSC_ACCESS_TOKEN_URI=https://your-instance.my.rubrik.com/api/client_token
RSC_BASE_URL=https://your-instance.my.rubrik.com

# Target CDM version to validate against
TARGET_CDM_VERSION=9.5
```

Optional Settings

Setting	Description	Default
CDM_CLUSTER_IDS	Comma-separated cluster UUIDs to assess (blank = all)	All clusters
CDM_CLUSTER_NAMES	Comma-separated cluster names to assess (blank = all)	All clusters
SKIP_DISCONNECTED_CLUSTERS	Skip disconnected clusters	true
LOG_LEVEL	Console log verbosity (INFO or DEBUG)	INFO
REPORT_FORMATS	Report formats to generate	csv,json,html

6. Running the Assessment

macOS / Linux

```
./run.sh
```

Windows

```
.venv\Scripts\Activate.ps1
python main.py
```

The assessment typically takes 2-3 minutes for 10 clusters. The HTML dashboard opens automatically when complete (macOS/Linux).

7. Understanding the Output

Reports are generated in the output directory:

Path	Contents
html/00_dashboard.html	Interactive multi-cluster overview dashboard

html/cluster_*.html	Per-cluster detail reports with sortable tables
csv/	Spreadsheet-compatible data for each assessment section
json/	Machine-readable data for programmatic analysis
assessment_*.log	Full debug log for troubleshooting

Report Sections Per Cluster

Section	Description
Cluster Identity & Health	Version, status, nodes, capacity, EOS status
Node Hardware & Disks	Node count, disk health, system status
Hypervisor Versions	vCenter, ESXi, Hyper-V SCVMM, Nutanix AHV versions
Database Platforms	MSSQL, Oracle, PostgreSQL, SAP HANA, MySQL, MongoDB, Db2
Physical Hosts	Windows and Linux hosts with OS version
NAS Systems	Vendor type, OS version, protocol support
Additional Workloads	Exchange (with version), AD, Kubernetes
Managed Volumes	Regular and SLA-based MVs, live mount status
SLA Domain Policies	Replication, archival, retention lock
Upgrade Pre-Checks	EOS status, upgrade paths, recommended version
Upgrade Blockers	Live mounts, running jobs, system status, DNS/NTP, service accounts
Compatibility Validation	All versions validated against target CDM matrix

8. Sending Results to Your Rubrik SE

After the assessment completes, a zip file is automatically created:

```
assessment_results_YYYYMMDD_HHMMSS.zip
```

Send this zip file to your Rubrik SE for review. It contains the HTML dashboard, CSV data, JSON data, and debug log.

Privacy

The output contains cluster names, versions, workload inventory, and compatibility findings. It does NOT contain:

- Credentials or secrets
- Backup data content
- User personal information
- Network topology details beyond cluster node IPs

9. Manual Verification Checklist

Some items cannot be checked via API and are flagged in the report for manual verification.

Before Upgrading to CDM 9.4+

☐ Check CDM UI → Settings → API Tokens → Remove legacy tokens

☐ Check CDM UI → Settings → Users → Service Accounts → Note CDM-native accounts

Before Upgrading to CDM 9.5.1+

☐ REQUIRED: Remove ALL CDM API tokens (RBK10900073)

☐ REQUIRED: Remove ALL CDM service accounts (RBK10900072)

☐ Migrate all automation to RSC service accounts

☐ Verify Azure cloud cluster nodes use Premium SSD v2 (if applicable)

Before Any Upgrade

☐ Review all BLOCKER findings in the report

☐ Dismount any active live mounts (VMware, MSSQL, Oracle)

☐ Unexport any managed volume exports

☐ Allow running backup jobs to complete

☐ Verify RSC connectivity is healthy

☐ Verify DNS and NTP are configured correctly

☐ Consider enabling support tunnel for Rubrik support access

10. What the Tool Checks

Upgrade Blockers (Critical)

Check	Severity	Method
CDM API Tokens	BLOCKER (9.5.1+)	Manual — CDM blocks API access by design
CDM Service Accounts	BLOCKER (9.5.1+)	Manual — CDM blocks API access by design
Active Live Mounts	BLOCKER	CDM Direct API
Running/Queued Jobs	WARNING	RSC GraphQL
RSC Connectivity	BLOCKER if failed	RSC GraphQL
System Status	BLOCKER if not OK	CDM Direct API

Version-Specific Warnings

Risk	Affected Versions	Severity
Nutanix AHV regressions	9.4.x (upgrade from 9.3)	WARNING
Oracle RAC ORA-01882	9.4.1, 9.4.2	WARNING

AD backup hanging	9.4.3	WARNING
AWS S3 archival issues	9.4.1+	INFO
Cloud cluster SSD v2	9.5+ (Azure)	WARNING
NAS relic visibility	9.4+	INFO

Compatibility Matrix Validation

The tool validates all detected versions against the Rubrik Compatibility Matrix:

Component	What Is Validated
VMware vSphere	vCenter and ESXi versions
Hyper-V	SCVMM version (incl. build number mapping)
Nutanix AHV	AOS version
Windows Server	Host OS version for RBS compatibility
Linux	RHEL, Ubuntu, CentOS, SLES, Debian versions
MSSQL	SQL Server version (2012-2022)
Oracle	Oracle version (11g-23ai)

11. Troubleshooting

Issue	Solution
Python not found	macOS: brew install python@3.11 Linux: sudo apt install python3 Windows: Download from python.org
SSL certificate warning	Cosmetic only — does not affect functionality. Fix: pip install pyOpenSSL
Authentication failed	Verify RSC_CLIENT_ID and RSC_CLIENT_SECRET in .env. Ensure service account is active.
CDM Direct API not available	Expected for cloud clusters with private IPs. Tool falls back to GraphQL automatically.
No clusters found	Verify RSC_BASE_URL. Check if CDM_CLUSTER_NAMES filter is set. Ensure service account has cluster read permissions.
Debug mode	Set LOG_LEVEL=DEBUG in .env. Check logs/ folder for details.

12. FAQ

Q: How long does the assessment take?

A: Typically 2-3 minutes for 10 clusters. Each cluster takes 10-15 seconds to assess.

Q: Does the tool make any changes to my clusters?

A: No. The tool is completely read-only. It only queries data — it never modifies configurations, policies, or workloads.

Q: Does the output contain sensitive data?

A: The output contains cluster names, versions, and workload inventory. It does NOT contain credentials, backup data, or user personal information.

Q: Why do some clusters show "CDM Direct API: Not available"?

A: Cloud clusters (AWS, Azure) use private IP addresses that are not reachable from your machine. The tool falls back to RSC GraphQL for these clusters automatically.

Q: Why does the tool flag CDM service accounts and API tokens for manual check?

A: The CDM REST API blocks programmatic access to these endpoints for ALL token-based authentication by design. This is a CDM framework-level protection, not a permissions issue. Manual verification via the CDM UI is required.

Q: Can I assess only specific clusters?

A: Yes. Set CDM_CLUSTER_NAMES or CDM_CLUSTER_IDS in the .env file to filter which clusters are assessed.

Q: What if I get zero findings?

A: Zero findings means no compatibility issues were detected for the target CDM version. This is a good result! Still complete the manual verification checklist before upgrading.